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**DEPARTMENT OF INFORMATION
TECHNOLOGY**

EXPLORING AND MANAGING THREADS IN WINDOWS OPERATING SYSTEM

Subject:

Operating Systems

Type:

Individual Performance Task

Platform:

Windows 11

Activity Title:

Hands-On Thread Exploration and Management in Windows

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I. Part A – Prepare the Windows Environment

Information	Details
Name:	Mary Anne P. Racelis
Section:	BSCS-3A
Computer Name:	LAPTOP-I0TC3NQG
Windows Version:	Windows 11
Date:	October 2, 2026
Start Time:	9:32 AM

Required Evidence 1

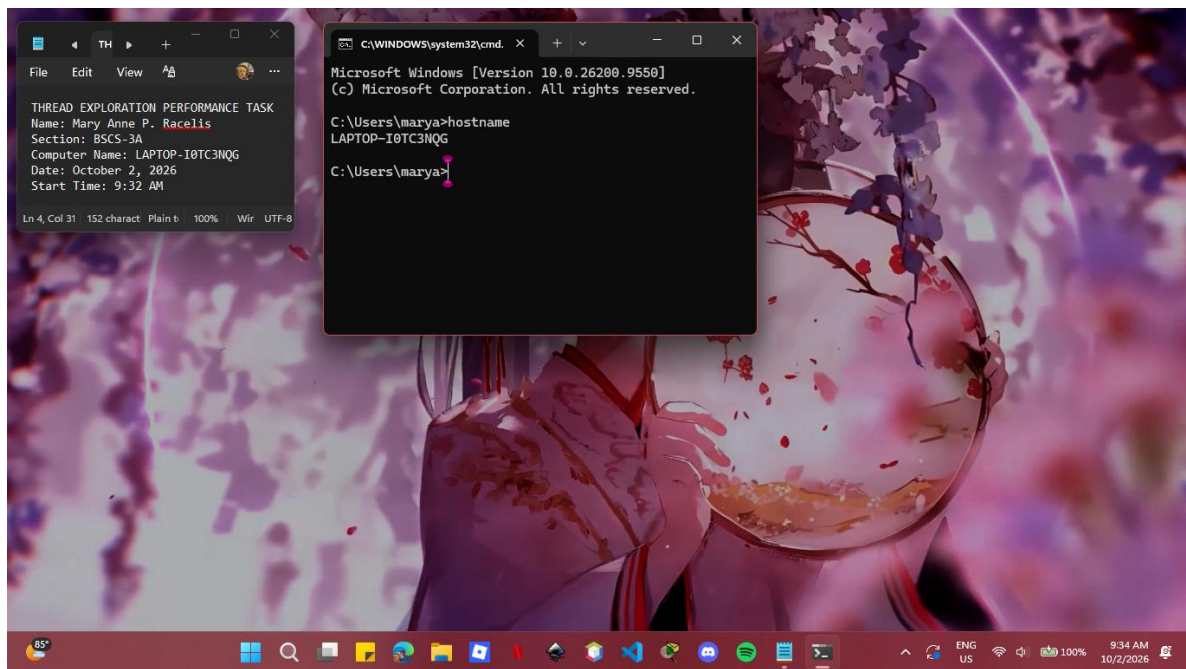


Figure 1. Evidence01_ LAPTOP-I0TC3NQG.png

II. Part C – Display the Threads Column

Required Evidence 2

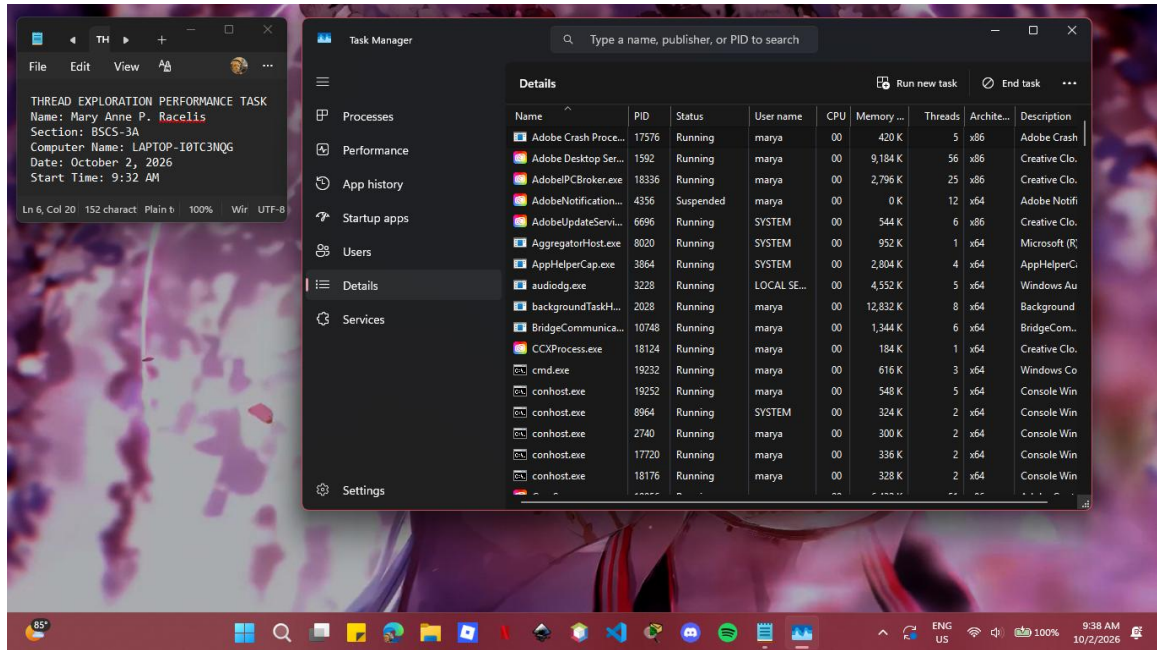
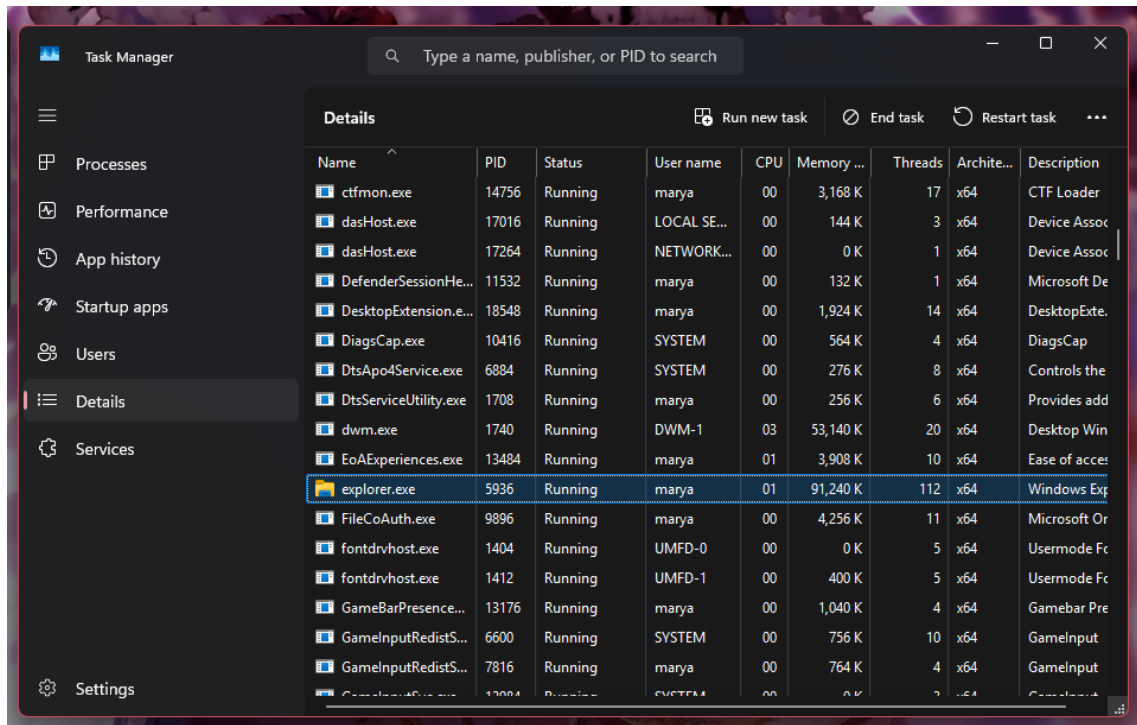


Figure 1. Evidence02_ThreadColumn.png

III. Part D – Collect Actual Thread Data

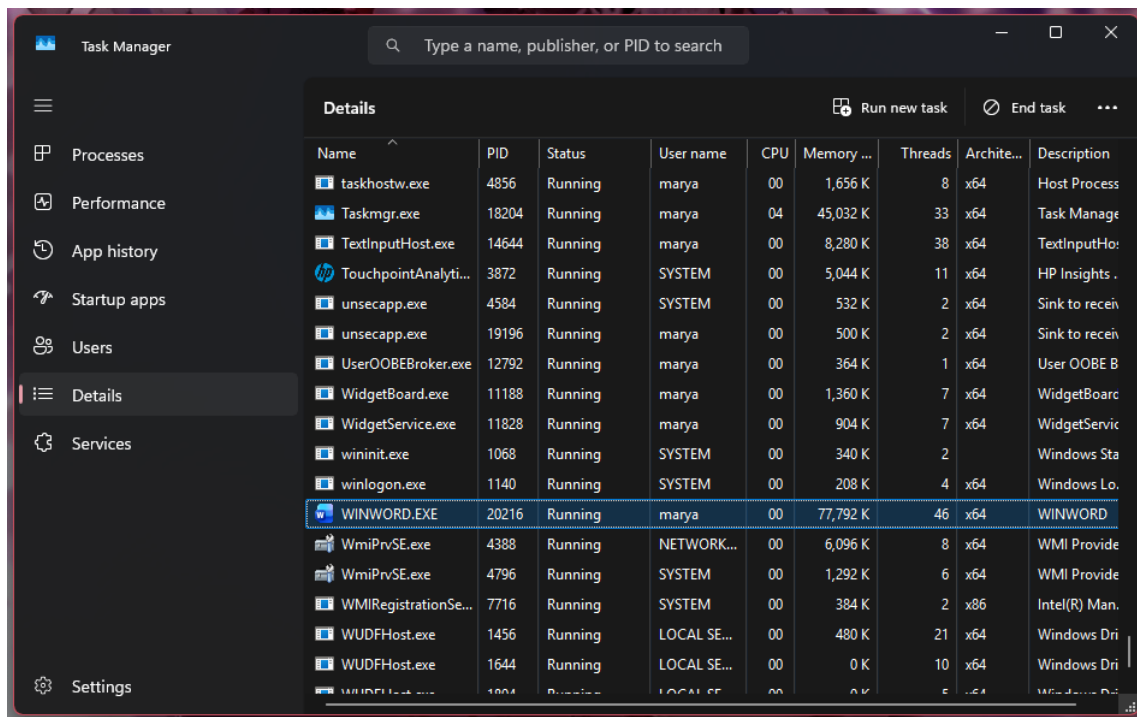
Process	PID	Thread Count	CPU	Memory
Browser (Edge)	18424	13	00	2,924K
Notepad	8172	21	00	6,892K
File Explorer	5936	112	01	91,240K
Calculator	18500	25	00	0K
Other (MS Word)	20216	46	00	77,792K

Required Evidence 3



Name	PID	Status	User name	CPU	Memory ...	Threads	Archite...	Description
ctfmon.exe	14756	Running	marya	00	3,168 K	17	x64	CTF Loader
dasHost.exe	17016	Running	LOCAL SE...	00	144 K	3	x64	Device Assoc
dasHost.exe	17264	Running	NETWORK...	00	0 K	1	x64	Device Assoc
DefenderSessionHe...	11532	Running	marya	00	132 K	1	x64	Microsoft De
DesktopExtension.e...	18548	Running	marya	00	1,924 K	14	x64	DesktopExte.
DiagsCap.exe	10416	Running	SYSTEM	00	564 K	4	x64	DiagsCap
DtsApo4Service.exe	6884	Running	SYSTEM	00	276 K	8	x64	Controls the
DtsServiceUtility.exe	1708	Running	marya	00	256 K	6	x64	Provides add
dwm.exe	1740	Running	DWM-1	03	53,140 K	20	x64	Desktop Win
EoAExperiences.exe	13484	Running	marya	01	3,908 K	10	x64	Ease of acce
explorer.exe	5936	Running	marya	01	91,240 K	112	x64	Windows Exp
FileCoAuth.exe	9896	Running	marya	00	4,256 K	11	x64	Microsoft Or
fontdrvhost.exe	1404	Running	UMFD-0	00	0 K	5	x64	Usermode Fc
fontdrvhost.exe	1412	Running	UMFD-1	00	400 K	5	x64	Usermode Fc
GameBarPresence...	13176	Running	marya	00	1,040 K	4	x64	Gamebar Pre
GameInputRedistS...	6600	Running	SYSTEM	00	756 K	10	x64	GameInput
GameInputRedistS...	7816	Running	marya	00	764 K	4	x64	GameInput
GameInputRedistS...	13004	Running	SYSTEM	00	0 K	2	x64	GameInput

Figure 1. Evidence03_ProcessThreads (1).png



Name	PID	Status	User name	CPU	Memory ...	Threads	Archite...	Description
taskhostw.exe	4856	Running	marya	00	1,656 K	8	x64	Host Process
Taskmgr.exe	18204	Running	marya	04	45,032 K	33	x64	Task Manage
TextInputHost.exe	14644	Running	marya	00	8,280 K	38	x64	TextInputHo:
TouchpointAnalyti...	3872	Running	SYSTEM	00	5,044 K	11	x64	HP Insights .
unsecapp.exe	4584	Running	SYSTEM	00	532 K	2	x64	Sink to recei
unsecapp.exe	19196	Running	marya	00	500 K	2	x64	Sink to recei
UserOOBEBroker.exe	12792	Running	marya	00	364 K	1	x64	User OOBEB
WidgetBoard.exe	11188	Running	marya	00	1,360 K	7	x64	WidgetBoarc
WidgetService.exe	11828	Running	marya	00	904 K	7	x64	WidgetServic
wininit.exe	1068	Running	SYSTEM	00	340 K	2		Windows Sta
winlogon.exe	1140	Running	SYSTEM	00	208 K	4	x64	Windows Lo.
WINWORD.EXE	20216	Running	marya	00	77,792 K	46	x64	WINWORD
WmiPrivSE.exe	4388	Running	NETWORK...	00	6,096 K	8	x64	WMI Provide
WmiPrivSE.exe	4796	Running	SYSTEM	00	1,292 K	6	x64	WMI Provide
WMIRegistrationSe...	7716	Running	SYSTEM	00	384 K	2	x86	Intel(R) Man.
WUDFHost.exe	1456	Running	LOCAL SE...	00	480 K	21	x64	Windows Dri
WUDFHost.exe	1644	Running	LOCAL SE...	00	0 K	10	x64	Windows Dri
WUDFHost.exe	1804	Running	LOCAL SE...	00	0 K	5	x64	Windows Dri

Figure 2. Evidence03_ProcessThreads (2).png

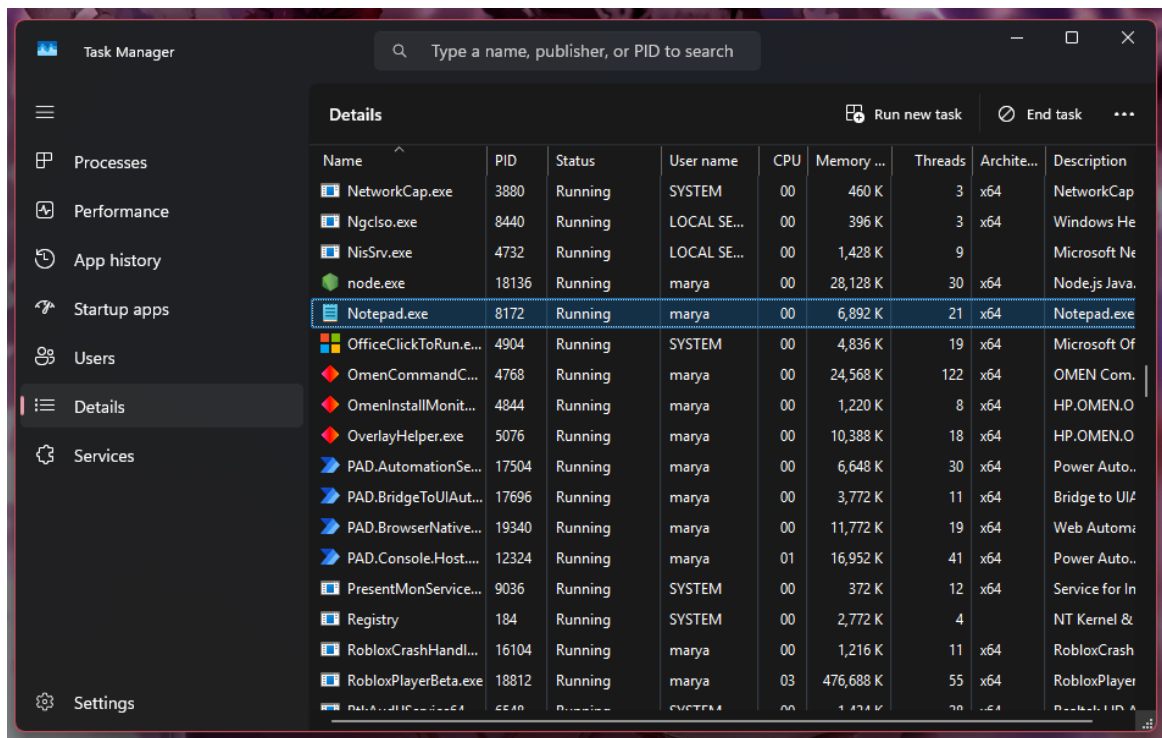


Figure 3. Evidence03_ProcessThreads (3).png

IV. Part E – Observe Thread Changes in a Web Browser

Observation	Browser Processes	Selected PID	Threads	CPU
A. One idle tab	11	5412	19	00
B. Five tabs	30	3936	12	00
C. Active browsing	31	20428	14	00



ire 1

V. Part G – Count Threads Using PowerShell

Information	Task Manager	PowerShell
Process Name	WINWORD.EXE	WINWORD
PID	20216	20216
Number of Threads	46	46

Required Evidence 5

The figure consists of two screenshots. The top screenshot shows the Windows Task Manager 'Details' tab. The 'WINWORD.EXE' process is selected, showing it is running with PID 20216, owned by 'marya', and has 46 threads. The bottom screenshot shows a Windows PowerShell terminal window. It displays the output of the command `Get-Process WINWORD`, which lists various system processes. Below this, it shows the output of `(Get-Process WINWORD).Threads.Count`, which returns 44. Then, it shows the output of `(Get-Process -id 20216).Threads.Count`, which returns 44. Finally, it shows the output of `(Get-Process -id 20216).Threads.Count` again, which returns 46.

Task Manager Details:

Name	PID	Status	User name	CPU	Memory	Threads	Archite...	Description
TouchpointAnalytics...	3872	Running	SYSTEM	00	3,616 K	13	x64	HP Insight
unsecapp.exe	4584	Running	SYSTEM	00	572 K	3	x64	Sink to rec
unsecapp.exe	19196	Running	marya	00	504 K	3	x64	Sink to rec
UserOOBEBroker.exe	12782	Running	marya	00	20 K	1	x64	User OOB
WidgetBoard.exe	11188	Running	marya	00	768 K	5	x64	WidgetBo
WidgetService.exe	11828	Running	marya	00	688 K	7	x64	WidgetSer
WindowsTerminal.exe	7660	Running	marya	00	12,820 K	18	x64	Windows T
wininit.exe	1068	Running	SYSTEM	00	0 K	1		Windows S
winlogon.exe	1140	Running	SYSTEM	00	200 K	4	x64	Windows L
WINWORD.EXE	20216	Running	marya	00	63,744 K	46	x64	WINWORD
WmiPrvSE.exe	19672	Running	NETWORK...	02	5,584 K	8	x64	WMI Provi
WmiPrvSE.exe	4796	Running	SYSTEM	00	804 K	6	x64	WMI Provi
WMIRegistrationServ...	7716	Running	SYSTEM	00	0 K	2	x86	Intel(R) Me
WUDFHost.exe	1456	Running	LOCAL SE...	00	480 K	21	x64	Windows [
WUDFHost.exe	1644	Running	LOCAL SE...	00	0 K	10	x64	Windows I
WUDFHost.exe	1804	Running	LOCAL SE...	00	0 K	5	x64	Windows I
XboxPcAppFT.exe	17968	Running	marya	00	1,328 K	7	x64	Xbox App

PowerShell Output:

```

PS C:\WINDOWS\system32> Get-Process WINWORD
Handles NPM(K) PM(K) WS(K) CPU(s) Id SI ProcessName
-----
2119 65 244164 150980 328.08 20216 1 WINWORD

PS C:\WINDOWS\system32> (Get-Process WINWORD).Threads.Count
44
PS C:\WINDOWS\system32> (Get-Process -id 20216).Threads.Count
44
PS C:\WINDOWS\system32> (Get-Process -id 20216).Threads.Count
56
PS C:\WINDOWS\system32> (Get-Process -id 20216).Threads.Count
50
PS C:\WINDOWS\system32> (Get-Process -id 20216).Threads.Count
46
PS C:\WINDOWS\system32>

```

Figure 1-2. Evidence05_TaskManagerVsPowerShell.png

VI. Part H – Explore Individual Threads

Selected Process: WINWORD

PID: 19056

Thread ID	Thread State	Wait Reason	Start Time	Priority Level
19052	Wait	UserRequest	10/2/2026 2:56:27 PM	Normal
19064	Wait	EventPairLow	10/2/2026 2:56:27 PM	Normal

Required Evidence 6

The screenshot displays two windows. On the left is the 'THREAD EXPLORATION PERFORMANCE TASK' window, showing details for a thread named 'Mary Anne P. Racelis' with section 'BSCS-3A' on computer 'LAPTOP-I0TC3INQG' dated 'October 2, 2026' at '9:32 AM'. On the right is a Windows PowerShell window showing the command `(Get-Process -Id 19056).Threads` and its output. The output lists two threads with their respective properties.

```
PS C:\WINDOWS\system32> (Get-Process -Id 19056).Threads

BasePriority      : 8
CurrentPriority   : 9
Id               : 19052
IdealProcessor    : 
PriorityBoostEnabled : True
PriorityLevel     : Normal
PrivilegedProcessorTime : 00:00:10.2343750
StartAddress     : 0
StartTime        : 10/2/2026 2:56:27 PM
ThreadState      : Wait
TotalProcessorTime : 00:00:50.0312500
UserProcessorTime : 00:00:39.7968750
WaitReason       : UserRequest
ProcessorAffinity : 
Site             : 
Container        : 

BasePriority      : 8
CurrentPriority   : 8
Id               : 19064
IdealProcessor    : 
PriorityBoostEnabled : True
PriorityLevel     : Normal
PrivilegedProcessorTime : 00:00:00.2031250
StartAddress     : 0
StartTime        : 10/2/2026 2:56:27 PM
ThreadState      : Wait
```

Figure 1. Evidence06_IndividualThreads.png

VII. Part I – Observe Thread States

Thread ID	Thread State	Wait Reason
11120	Running	
11016	Wait	UserRequest
10592	Wait	Unknown
8444	Wait	Unknown
14300	Wait	UserRequest

Required Evidence 7

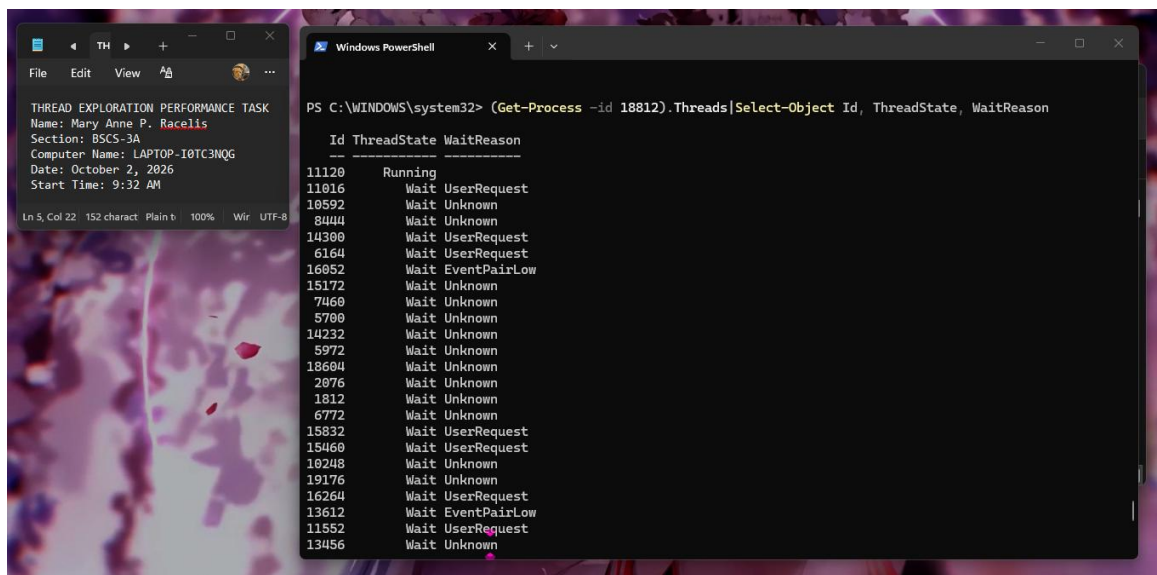


Figure 1. Evidence07_ThreadStates.png

VIII. Part J – Perform A Live Thread Experiment

Process Name: Notepad.exe

PID: 8172

Condition	Threads	CPU	Memory
Idle	21	0%	1,340K
Active	36	0%	38,476K
After Activity	21	0%	4,124K

Required Evidence 8

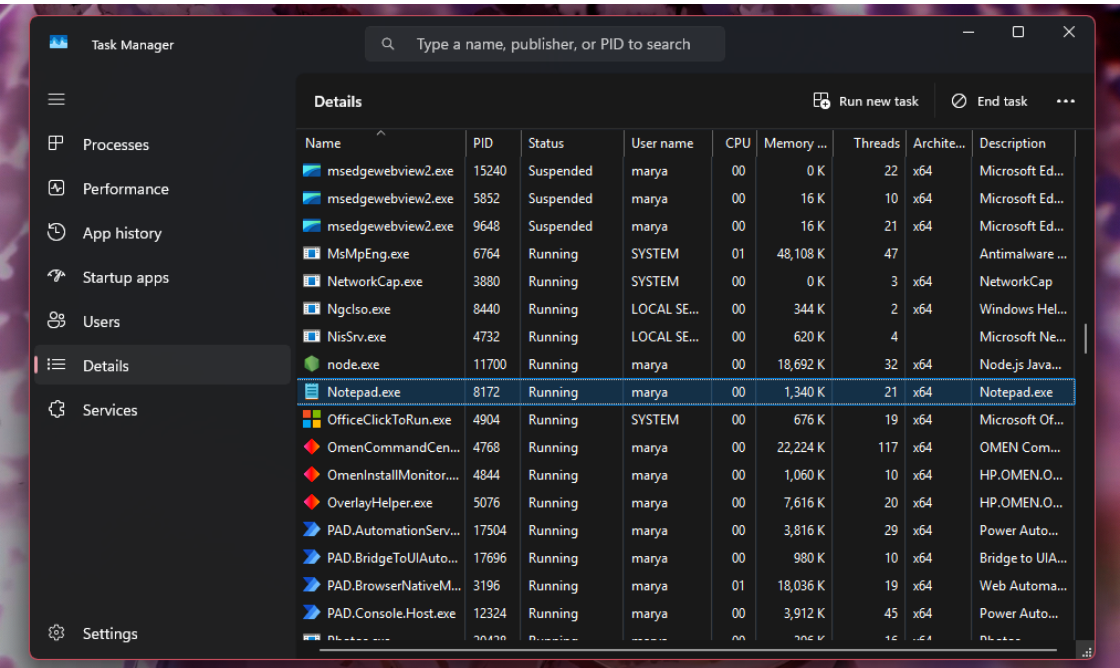


Figure 1. Evidence08_Idle.png

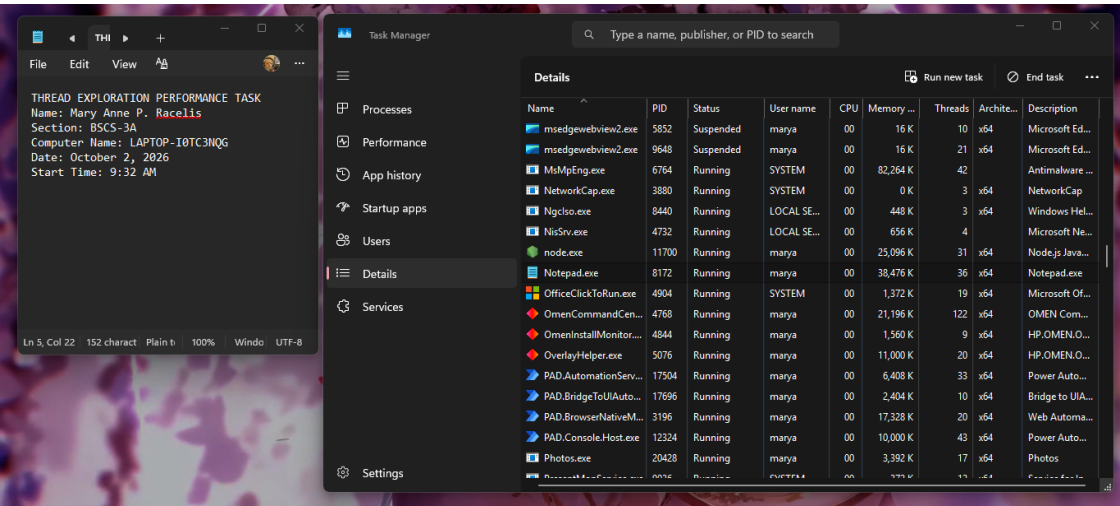


Figure 2. Evidence08_Active.png

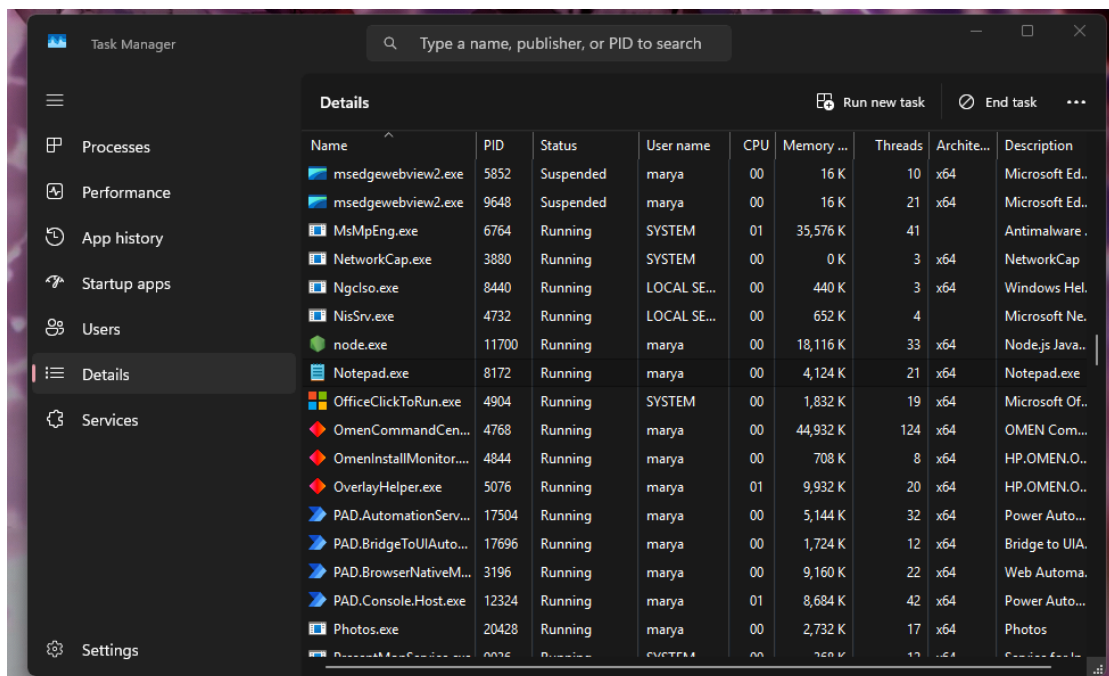


Figure 2. Evidence08_After.png

IX. Part L – Examine CPU Cores

Item	Actual Value
Processes	276
Threads	3742
Handles	139549
Cores	6
Logical Processors	8

Required Evidence 9

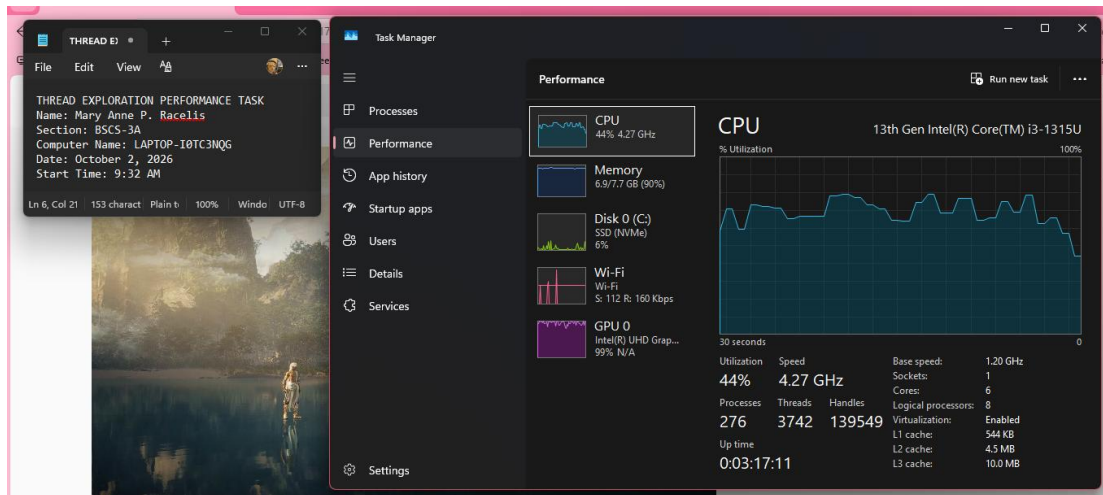


Figure 2. Evidence09_CPULoad.png

X. Part M – Visualize Logical Processor Activity

Required Evidence 10

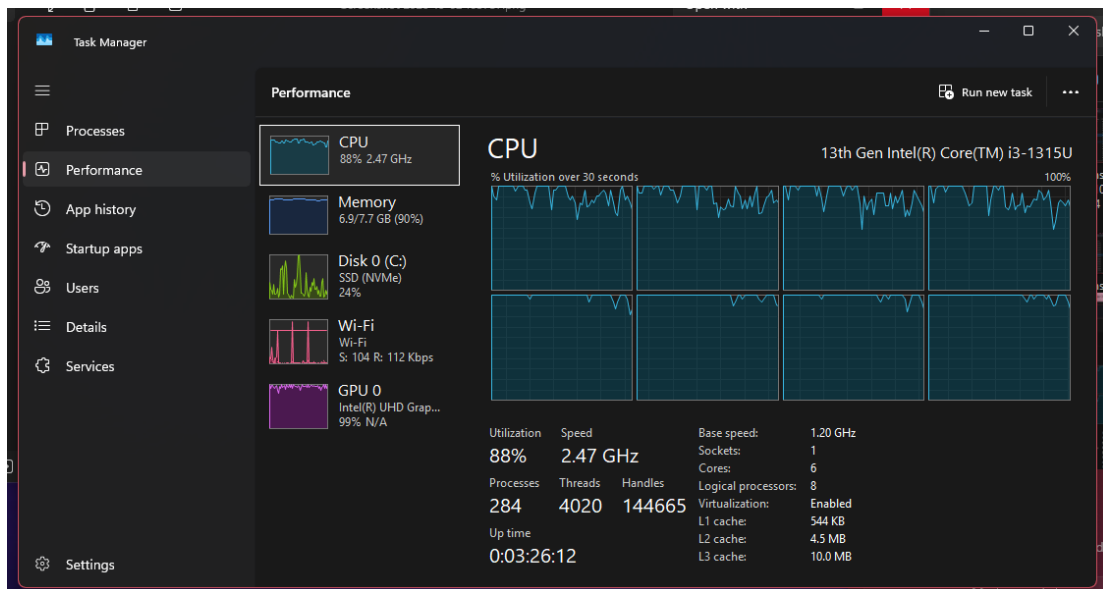


Figure 1. Evidence10_LogicalProcessors.png

XI. Part N – Process vs Thread Investigation

Application: Spotify.exe

PID: 19644

Number of Threads: 23

Number of CPU cores: 6

Number of logical processors: 8

Required Evidence 11

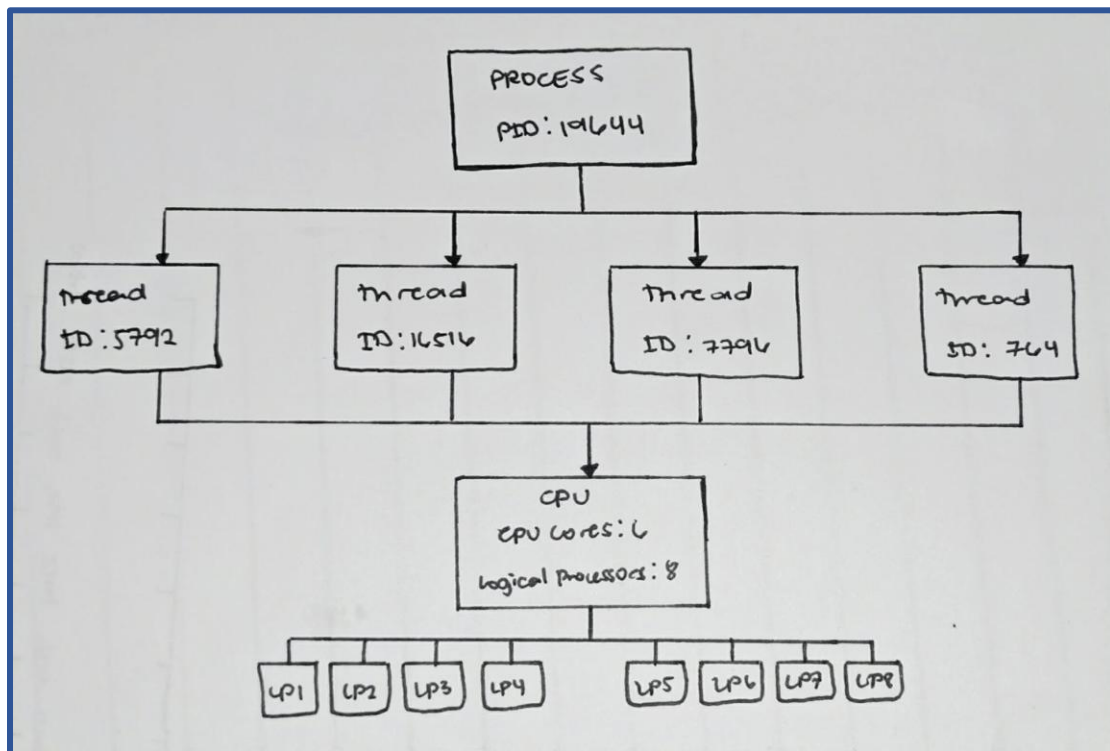


Figure 1. Evidence11_ProcessThreadDiagram.png

XII. Part O – Controlled Process Restart Experiment

Observation	PID	Threads
Before Closing	8172	23
After Closing	9096	26

Required Evidence 12

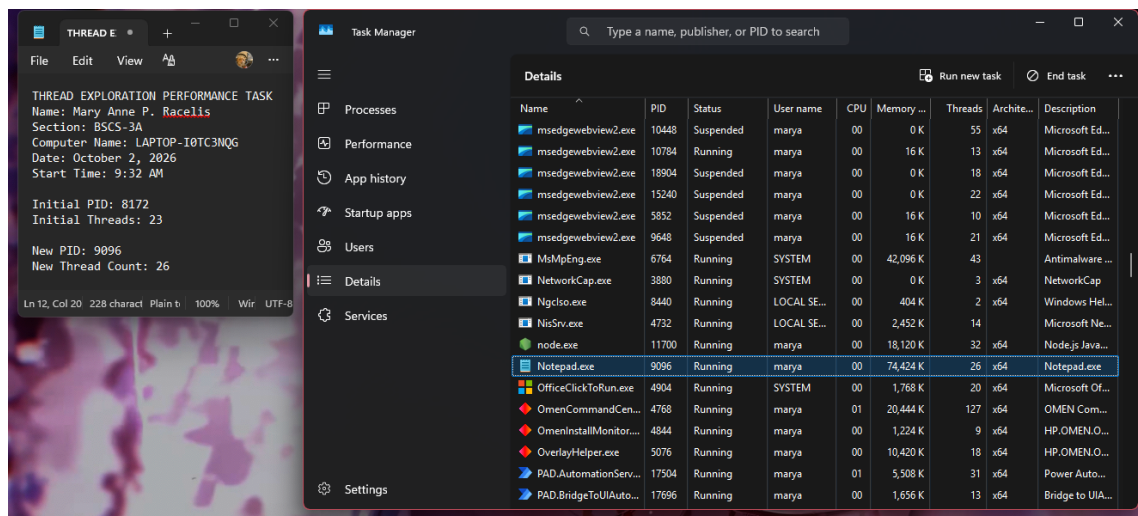


Figure 1. Evidence12_ProcessRestart.png

XIII. Practical Interpretation

On my computer, File Explorer had the highest thread count among the applications with 112 thread count. When I increased application activity by opening and using more applications, the CPU usage graphs became almost fully filled with blue, and Roblox started running slowly. However, my computer was still able to multitask without crashing, and the applications continued to work. The thread count did not remain constant because it changed depending on how much an application was being used. CPU utilization reached 88% and continued to increase as I increased application activity and opened more applications. My computer has 6 CPU cores and 8 logical processors. When I viewed the logical processor graphs while performing multiple tasks, I observed activity across the graphs, showing that my computer was using its logical processors while handling the increased workload. When I restarted an application, its PID changed, showing that Windows

created a new process application after the previous one was closed. I think Windows needs multiple threads so multiple applications can run at the same time and different tasks within an application can also make progress, such as searching and loading pages in a browser.